

TripGenie AI: An Intelligent Multi-Modal Travel Planning Framework Integrating Dense Semantic Retrieval-Augmented Generation (RAG) with Deterministic Financial Modeling

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Repository: github.com/sarapnayan-svg/TRIPGENIE-AI

Abstract—Commercial automated trip planning systems and generative Large Language Models (LLMs) suffer from three critical shortcomings: (1) **hallucinatory generation** of non-existent points of interest (POIs) or closed attractions, (2) **financial stochasticity**, wherein approximate travel costs lack mathematical rigor and fail to model physical constraints such as lodging room multipliers or vehicle capacities, and (3) **modal fragmentation**, requiring travelers to navigate disparate services for itineraries, live weather, geographic routing, and document exports.

To resolve these challenges, this paper presents **TripGenie AI**, an end-to-end, multi-modal travel intelligence platform. TripGenie AI implements an 8-stage Retrieval-Augmented Generation (RAG) pipeline utilizing dense semantic vector representations (all-MiniLM-L6-v2, 384 dimensions) and sub-millisecond in-memory matrix dot-product search with destination affinity boosting. Retrieved factual chunks are injected into Anthropic Claude 3.5 Sonnet to synthesize verified, hallucination-free itineraries. Concurrently, a decoupled deterministic financial engine models expenditures across five discrete categories (Accommodation, Food, Transit, Activities, and Contingency Reserve) with automated deficit resolution. Real-world spatial grounding is enforced via Haversine geodesic routing over OpenStreetMap coordinates, real-time meteorological observations via World Meteorological Organization (WMO) station telemetry, and zero-server client-side vector PDF dossier compilation. Experimental evaluations confirm that TripGenie AI reduces POI hallucination to 0% within indexed catalogs, achieves 100% mathematical consistency in budget allocation, and provides sub-50ms retrieval latency on commodity hardware.

Keywords: Retrieval-Augmented Generation (RAG), Large Language Models (LLM), Dense Vector Embeddings, Deterministic Financial Planning, Spatial Computing, Haversine Distance, Intelligent Tourism Systems.

I. INTRODUCTION

The integration of Generative Artificial Intelligence (GenAI) into e-tourism has attracted substantial interest due to the natural language reasoning capabilities of modern Large Language Models (LLMs) [1]. However, deploying unconstrained LLMs in real-world travel planning poses severe reliability hazards. LLMs generate text based on probabilistic token likelihoods derived from vast pre-training corpora, rather than factual spatial databases [2]. Consequently, commercial conversational agents frequently hallucinate non-existent attractions, suggest monuments with incorrect operating schedules, or recommend logically impossible day schedules spanning hundreds of kilometers without transit feasibility [3].

Furthermore, financial planning represents a notable limitation of autoregressive language models. Budget allocation requires deterministic arithmetic: calculating hotel room requirements based on party size, night counts, per-kilometer fuel tariffs, and ticket costs. When prompted for travel budgets, general-purpose LLMs generate broad approximations that ignore party scaling laws, resulting in severe budgetary deficits during actual travel [4].

Finally, modern travelers encounter extreme cognitive load caused by application fragmentation. A typical traveler must consult an LLM for inspiration, an online travel agency (OTA) for accommodation pricing, meteorological platforms for weather forecasts, a mapping service for geographic waypoints, and manual spreadsheet utilities for expense budgeting [5].

To address these systemic shortcomings, this paper presents **TripGenie AI**, characterized by five major contributions: (1) an 8-stage RAG semantic search engine with destination affinity score boosting; (2) a deterministic 5-category financial calculation engine; (3) spatial Haversine geodesic route modeling with verified GPS coordinates; (4) live WMO atmospheric telemetry integration; and (5) a client-side vector PDF compiler for zero-server document portability.

II. RELATED WORK

A. Generative LLMs in Domain-Specific Planning: Early automated travel recommenders relied heavily on collaborative filtering and rule-based heuristics [6]. With Transformer architectures [7], conversational travel assistants transitioned to generative models. While models such as GPT-4 and Claude exhibit conversational fluency, studies by Zhang et al. [8] indicate that up to 34% of travel itineraries generated by ungrounded LLMs contain factual inaccuracies regarding opening hours, entry fees, or spatial proximity.

B. Retrieval-Augmented Generation (RAG): Formalized by Lewis et al. [9], RAG combines parametric model memory with non-parametric external memory. By retrieving top-k relevant text chunks at inference time, RAG models significantly minimize factual hallucination in knowledge-intensive tasks [10]. In domain-specific travel applications, semantic vector search ensures that generative prompts are constrained to authentic, verified documents.

III. THEORETICAL FRAMEWORK & MATHEMATICAL FORMULATION

A. Dense Semantic Vector Normalization: Let $D = \{d_1, d_2, \dots, d_N\}$ represent the corpus of curated travel document chunks. Each document and user query is mapped into R^{384} dense vector space via SentenceTransformer E. Euclidean L2 normalization is applied to achieve unit length:

$$\mathbf{v_hat} = \mathbf{v} / ||\mathbf{v}||_2 = \mathbf{v} / \sqrt{\sum(\mathbf{v_j}^2)}, \quad \mathbf{q_hat} = \mathbf{q} / ||\mathbf{q}||_2$$

Because vectors possess unit length, the cosine similarity between query q and document chunk d_i reduces strictly to the vector dot product, executable in sub-millisecond matrix BLAS operations:

$$\text{Sim_cos}(\mathbf{q}, \mathbf{d_i}) = \mathbf{q_hat} \cdot \mathbf{v_hat_i} = \sum(\mathbf{q_hat_j} * \mathbf{v_hat_ij})$$

B. Spatial Affinity & Intent Boosting: To ensure absolute geographical relevance and prevent cross-regional bleed (e.g. matching beach chunks from Goa when planning Kerala), a structured relevance adjustment delta is incorporated:

$$\mathbf{S_final}(\mathbf{q}, \mathbf{d_i}) = \text{Sim_cos}(\mathbf{q}, \mathbf{d_i}) + \text{delta_dest}(\mathbf{d_i}) + \text{delta_cat}(\mathbf{d_i})$$

Where $\text{delta_dest} = +0.15$ for exact destination matches, and $\text{delta_cat} = +0.05$ for user interest matches. Candidate chunks are prioritized by $\mathbf{S_final}$.

C. Deterministic Financial Formulation: Total expenditure C_{total} is partitioned across five discrete categories:

$$\mathbf{C_total} = \mathbf{C_acc} + \mathbf{C_food} + \mathbf{C_trans} + \mathbf{C_act} + \mathbf{C_misc}$$

- Accommodation: $C_{\text{acc}} = \text{ceil}(T / 2) * (\text{Days} - 1) * P_{\text{tier}}$, where T is traveler count and P_{tier} is nightly rate.
- Food Allowance: $C_{\text{food}} = T * \text{Days} * F_{\text{style}}$, where F_{style} is per-person daily dining tier.
- Local Transportation: $C_{\text{trans}} = \text{Days} * (\text{ceil}(T / V_{\text{cap}}) * K_{\text{rental}} + F_{\text{fuel}})$.
- Curated Activities: $C_{\text{act}} = T * \sum(\text{Tariff_selected})$.
- Contingency Reserve: $C_{\text{misc}} = 0.07 * (C_{\text{acc}} + C_{\text{food}} + C_{\text{trans}} + C_{\text{act}})$.

D. Haversine Spatial Geodesic Distance: Real-world distance between consecutive POIs (ϕ_1, λ_1) and (ϕ_2, λ_2) is computed over a spherical Earth model ($R_E = 6,371$ km):

$$\mathbf{a} = \sin^2(\Delta\phi / 2) + \cos(\phi_1) * \cos(\phi_2) * \sin^2(\Delta\lambda / 2)$$

$$\mathbf{d_geodesic} = 2 * R_E * \arcsin(\sqrt{\mathbf{a}}), \quad \mathbf{d_road} \approx 1.30 * \mathbf{d_geodesic}$$

IV. SYSTEM ARCHITECTURE & IMPLEMENTATION

TripGenie AI is architected as a decoupled, multi-modal microservices system comprising five operational layers:

Layer	Component Stack	Primary Functionality
Client (SPA)	React 19, Vite 5, Tailwind CSS, Leaflet.js, jsPDF	Interactive timeline, visual budget charts, GPS maps, client-side vector PDF generation
API & Security	FastAPI, Uvicorn, Pydantic v2, PyJWT, Bcrypt	RESTful endpoints, request validation, CORS control, JWT session authentication
RAG Pipeline	Sentence-Transformers (all-MiniLM-L6-v2), NumPy	Query normalization, 384-dim dense embedding, cosine dot-product vector search
Domain Engines	Deterministic Budget Planner, Hotel Scorer, Spatial Engine	5-category budget calculation, room/night scaling, proximity scoring, Haversine routing
External / DB	Anthropic Claude 3.5 Sonnet, Open-Meteo API, SQLAlchemy 2	Natural language narrative synthesis, live meteorological telemetry, SQLite/PostgreSQL storage

Client-Side Vector PDF Engine: To eliminate server rendering bottlenecks, TripGenie AI implements pure in-browser PDF generation using jsPDF and jspdf-autotable. Table widths are calibrated to exact 182 mm margins, and dynamic two-pass canvas execution computes dynamic headers and footers ('Page X of Y') with Indian currency symbol ('₹') formatting.

V. EXPERIMENTAL EVALUATION & RESULTS

The system was evaluated over a standardized benchmark of 50 multi-day travel requests spanning Goa, Jaipur, Kerala, Manali, and Rishikesh across party sizes from 1 to 8 travelers and budgets from ₹10,000 to ₹250,000.

Table I: POI Hallucination Rate Comparison

Architecture Configuration	Tested Queries	Fabricated POIs	Inaccurate Tariffs	Hallucination Rate
Ungrounded Claude 3.5 Sonnet	50	14	22	28.0%
TripGenie AI (RAG Grounded)	50	0	0	0.0%

Table II: System Latency Profile

Pipeline Sub-Operation	Execution Platform	Average Latency (ms)
Vector Matrix Dot-Product Search	Local CPU (NumPy BLAS)	1.2 ms
WMO Weather Telemetry Query	Open-Meteo API	142.0 ms
Claude 3.5 Sonnet LLM Synthesis	Anthropic Messages API	2,180.0 ms
In-Browser PDF Dossier Compilation	Client-Side WebAssembly/V8	85.0 ms
Total End-to-End Latency	Full System Pipeline	~2.4 seconds

VI. DISCUSSION & ARCHITECTURAL TRADE-OFFS

In-Memory Matrix vs. External Vector DB: TripGenie AI utilizes an in-memory normalized NumPy vector matrix rather than an external vector database (such as Pinecone, Weaviate, or Milvus). For domain catalogs containing up to 10,000 document chunks, an in-memory matrix requires zero network serialization latency, avoids monthly SaaS hosting costs, and executes dot-product similarity in under 2 milliseconds. For enterprise scaling exceeding 100,000 chunks, migrating to pgvector or Qdrant represents the optimal evolutionary path.

Decoupled Deterministic Arithmetic: Rather than tasking the LLM with complex multi-step arithmetic in its hidden states, TripGenie AI pre-computes financial allocations deterministically in Python and passes the pre-calculated financial dictionary to the LLM prompt. This architectural separation guarantees mathematical precision while utilizing the LLM exclusively for semantic organization and narrative synthesis.

VII. CONCLUSION & FUTURE WORK

This paper presented **TripGenie AI**, an intelligent multi-modal travel planning framework resolving the primary challenges of generative AI in tourism. By coupling an 8-stage dense semantic RAG pipeline (all-MiniLM-L6-v2) with Anthropic Claude 3.5 Sonnet, TripGenie AI eliminates POI hallucinations within indexed catalogs. Its deterministic financial engine enforces mathematical precision across five budget categories, while spatial Haversine modeling, live WMO weather observations, and zero-server vector PDF compilation deliver an integrated, production-ready traveler experience.

Future enhancements will incorporate automated dynamic web scraping for live local festival schedules, direct integration with IRCTC and Amadeus APIs for real-time rail/flight ticketing, and WebSocket-driven multi-user collaborative group planning.

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